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# A Neoproterozoic Geomagnetic Field Reversal from the Kurnool Group, India: Implications for Stratigraphic Correlation and Formation of Gondwana

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**Abstract:** The Proterozoic Cuddapah Supergroup is predominantly of normal magnetization but has a reversely magnetized level coinciding with the Srisailam Quartzite at the top of the succession. The Kurnool Group overlies the Cuddapah Supergroup with an unconformity and we present palaeomagnetic results from the Kurnool Group members comprising the Banganapalli Quartzite and the Narji Limestone. The Neoproterozoic Banganapalli Quartzite carries stable normal and reverse magnetizations suggesting a geomagnetic field reversal at the time of its formation. The Narji Limestone is normally magnetized. A similar magnetization to that in the Banganapalli Quartzite is reported ~1000km away in the Baghain Sandstone of the Vindhyan Supergroup. A possible correlation is made between Cuddapah Supergroup and Kurnool Group with the Vindhyan Supergroup. Based on palaeomagnetic poles, palaeolatitudes, palaeoclimatic and structural evidence it is suggested that India and Australia collided possibly during Neo-proterozoic times in an attempt to form larger Gondwana.

**Keywords**: Palaeomagnetism, Reversal of poles, Magnetostratigraphy, Gondwana evolution, Cuddapah Supergroup, Kurnool Group, Andhra Pradesh.

## INTRODUCTION

The crescent shaped intracratonic Cuddapah basin of Peninsular India covers an area of 44,500 km$^2$ in Andhra Pradesh. It is of Proterozoic age and consists of the Cuddapah Supergroup overlain by the Kurnool Group. The Kurnool Group occurs in two separate outliers as the Kurnool basin in the central part and the Palnad basin in the northeastern part of the Cuddapah basin (Fig.1). The formations occurring within the Kurnool and Palnad basins are considered to be essentially coeval and are given the common assignment as the Kurnool Group (King, 1872; Krishnan, 1982, Nagaraja Rao et al. 1987). The stratigraphy of the Cuddapah Supergroup and the Kurnool Group is given in Table 1.

The first attempts at palaeomagnetic study of igneous intrusives into the Papaghni and Chitravati Groups (called "Cuddapah Traps") were not successful (Athavale et al. 1963). Another study of the same formations from a larger number of sites over a wider area also failed to yield useful data (Prasad, 1966). However, Prasad and Damodara Reddy (1972) reported stable directions of D/I = 294° /-6° and D/I = 295°/-10° from a thin band of sandstone in the Tadpatri Formation and metamorphosed shale respectively. Clark (1983) isolated a similar magnetization (D/I = 299°/-6°) from the Cuddapah Traps by stepwise thermal demagnetization. Detailed work was carried out by Prasad et al. (1987) on the Gulcheru Quartzite, Tadpatri Shale and Quartzite and the Bairenkonda Quartzite from the Papaghni, Chitravati and Nallamalai Groups respectively and the Srisailam Quartzite, which is the topmost formation of Cuddapah Supergroup. The Gulcheru, Tadpatri and Bairenkonda Formations yielded a mean normal direction of D/I = 298°/-7° while the Srisailam Quartzite had a mean reverse direction of D/I = 140° /0°. The aim of this work is to look for the continuation, or otherwise, of this reversal into the Kurnool Group by palaeomagnetic study of the Banganapalli Quartzite, which overlies Srisailam Quartzite with an unconformity and the Narji Limestone, which conformably overlies the Banganapalli Quartzite.

In the stratigraphic framework of India a classic

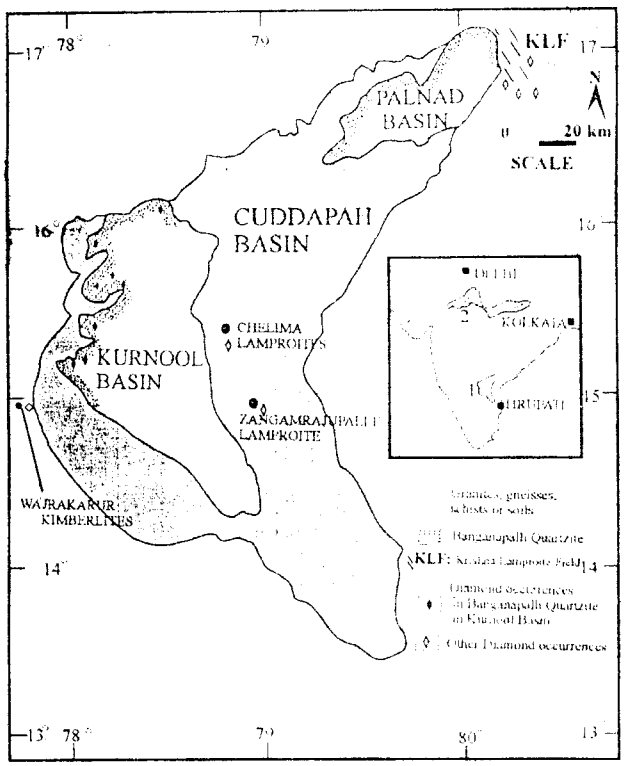



**Fig.1.** Outline map of Cuddapah basin showing Cuddapah, Kurnool and Palnad basins. *Inset map:* **1.** Cuddapah basin; **2.** Vindhyan basin (*after* Geological Survey of India, 1981).

**Table 1.** Stratigraphy of Cuddapah Supergroup and Kurnool Group (modified after Nagaraja Rao et al.1987)

| | Group | Formation |
|---|---|---|
| | Kurnool Group | Nandyal Shale |
| | | Koilkuntla Limestone |
| | | Paniam Quartzite |
| | | Owk Shale |
| | | Narji Limestone ** |
| | | Banganapalli Quartzite ** |
| | | ---------- Unconformity ---------- |
| CUDDAPAH SUPERGROUP | | Srisailam Quartzite * |
| | | ---------- Unconformity- ---------- |
| | Nallamalai Group | Cumbum (Pullampet) Shale |
| | | Bairenkonda (Nagari Quartzite) * |
| | | ---------- Angular Unconformity- ---------- |
| | Chitravati Group | Gandikota Quartzite |
| | | Tadpatri Shale and Quartzite with intrusives* |
| | | Pulivendla Quartzite |
| | | ---------- Disconformity- ---------- |
| | Papaghni Group | Vempalle Limestone and basic flows and intrusives |
| | | Gulcheru Quartzite * |
| | | ---------- Unconformity- ---------- |
| | | Archaean |

*Palaeomagnetically studied formations (Prasad et al. 1987);
** This work

correlation is recognized between the Kurnool Group and the Rewa and Bhander Groups of the Vindhyan Supergroup based on the presence of diamondiferous horizons in both sequences (Krishnan, 1982). A palaeomagnetic correlation of these groups is attempted here. Based on the palaeomagnetic poles and the palaeolatitude constraints, possible collision of India and Australia in the formation of Gondwana during the Proterozoic can also be evaluated.

## SAMPLING

The Kurnool Group of formations consists of conglomerate, quartzite, carbonate and shale facies. These are assigned to six formations and are thought to have formed by continuous and uninterrupted sedimentation (Nagaraja Rao et al. 1987). Sampling for this study is restricted to the Banganapalli Quartzite, which is the basal unit of the Kurnool Group, and the Narji Limestone which overlies it. The Banganapalli Quartzite is a well-known formation in the geology of India because it has a diamondiferous conglomeratic horizon at the base. The thickness ranges from 10-50 m and is composed of brown coloured sandstone, which is coarse-grained and gritty at lower levels. This gritty brown zone is 5 to 20 m thick and becomes gradually more uniform with a smaller grain size and lighter colour at higher levels. Sampling is restricted to the gritty layer and the brown coloured sandstone. A total of 106 samples from 9 sites in the Kurnool basin (Fig.2) and 97 samples from 11 sites in the Palnad basin (Fig.3) were collected. The Narji Limestone, which overlies Banganapalli Quartzite is a thicker formation (100–200 m) and was sampled at 3 sites over a lateral distance of about 10 km. The limestone is uniformly ferruginous and brown in colour. At each site oriented samples were collected with a minimum of 3 samples per site except at a few sites. Geological tilt at each site was noted and sampling details are given in Figs.2 and 3.

## PALAEOMAGNETISM

A total of 361 specimens prepared from 203 samples from the Banganapalli Quartzite in both the Kurnool basin (KB) and Palnad basin (PB) were measured for NRM directions. Forty three specimens from 19 samples of the

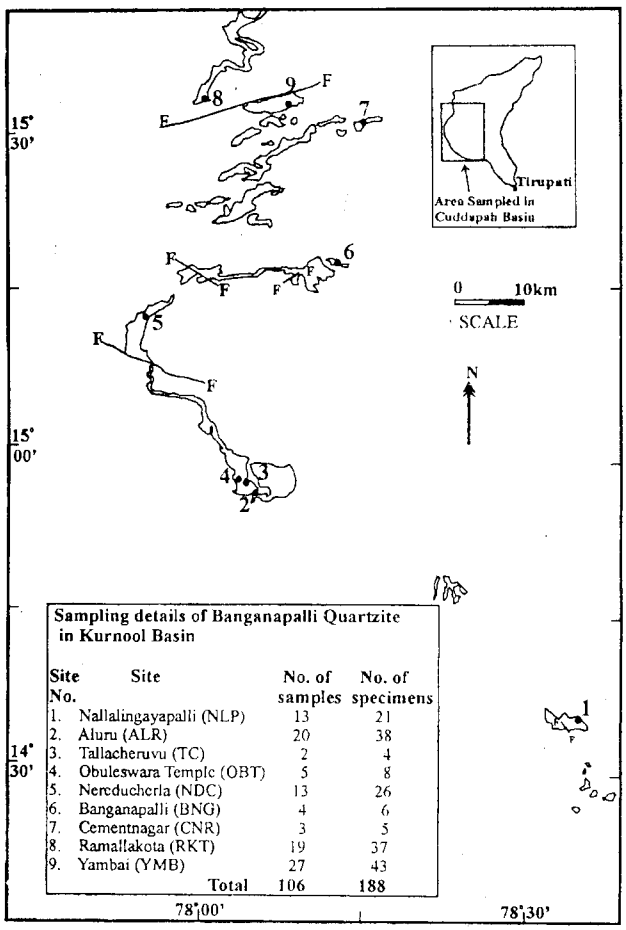


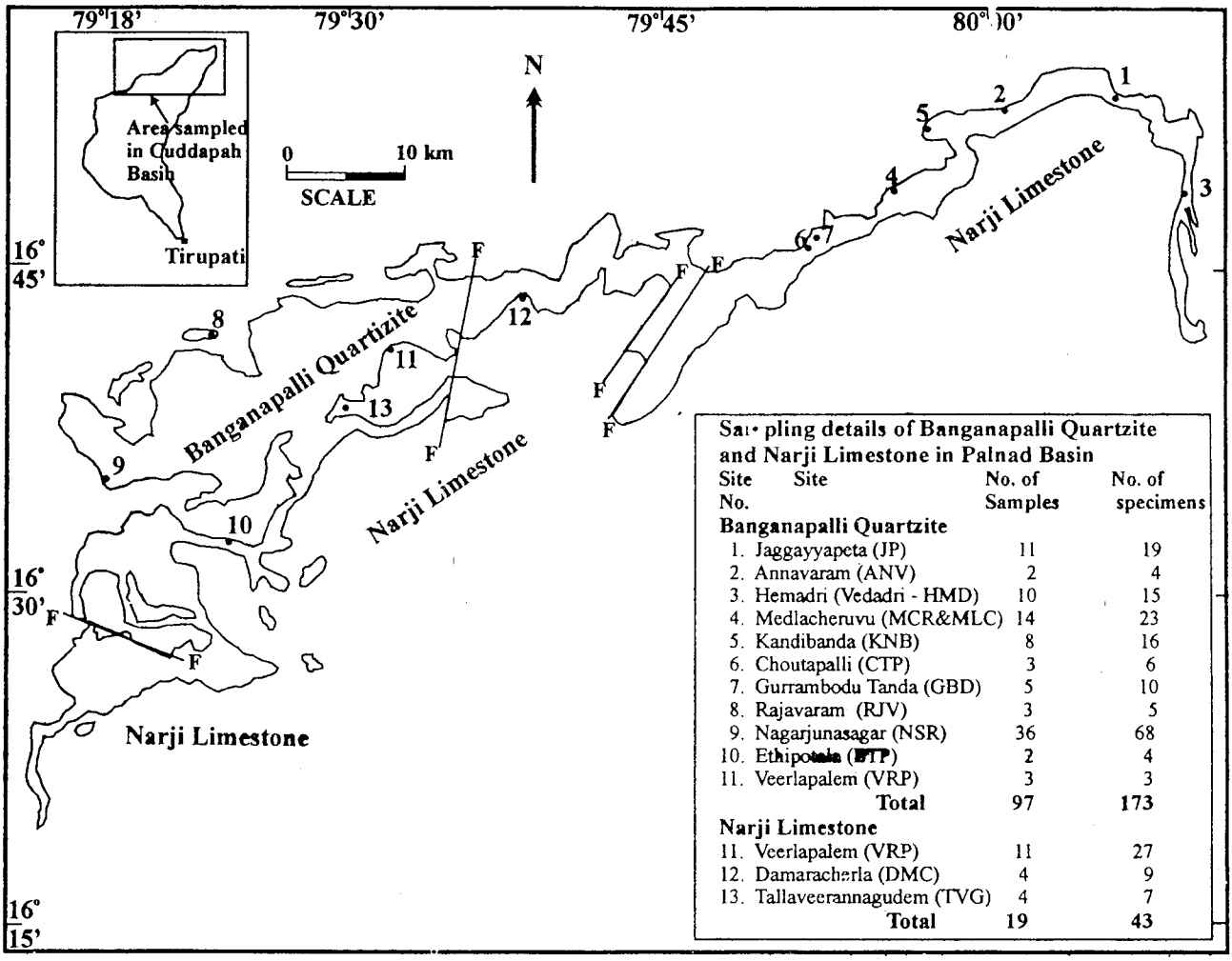

**Fig.2.** Outcrops of Banganapalli Quartzite in Kurnool basin. *Inset:* Outline map of Cuddapah basin showing the area sampled for palaeomagnetic study. Details of sampling is given in table as inset. Numbers in the Figure indicate site number given in table. F-F: Fault. (*after* Geological Survey of India, 1981)

Narji Limestone from the Palnad basin were also measured. The results showed that samples from the same site exhibit consistency of declination (D) and inclination (I) with some exceptions. An important character of the preliminary results from the Banganapalli Quartzite (BQ) is that it shows both normal and reversed directions while the Narji Limestone (NL) shows only normal directions. The Aluru site (ALR) from the KB shows 4 samples with reversed and 5 with normal directions. The Hemadri site (HMD) of the BQ from PB shows a very consistent reversed direction with negative inclination from all specimens from 10 samples. In this paper the terms "normal" and "reversed" are used by analogy with the direction of the present day geomagnetic field although it is not known with certainty that the normal directions and reversed directions correspond with the equivalent polarities of the present day geomagnetic field.

The sites, Nallalingayapalli (NLP), Nereducherla (NDC), Ramallakota (RKT) and Banganapalli (BNG) sites in the KB and the Jaggayyapet (JP), Medlacheruvu (MCR & MLC) and Nagarjunasagar (NSR) sites of the PB show general consistency of normal or reversed directions. The Damaracherla (DMC) and Veerlapalem (VRP) sites from the PB show consistent normal directions. The other sites from both basins show mostly inconsistent directions.

Thermal and alternating field demagnetization tests were carried out on BQ and NL specimens. A pilot thermal demagnetization, from room temperature to the Curie temperature in 8 steps at 100°C interval, of six selected specimens of BQ from both basins showed that magnetizations were generally subtracted and component behaviour recovered above 325°C (Fig.4). Therefore batch thermal demagnetization was carried out in steps beginning with the NRM at room temperature, then to 325°C, 425°C, 525°C, 600°C, 650°C and 680°C. At about 680°C the specimens loose all their magnetization indicating that the carrier of magnetization is hematite. Some specimens are initially very weak and loose much of their magnetization at lower temperatures of 300°C to 400°C. Such specimens are excluded from further consideration. Pilot alternating field demagnetization (AFD) indicated that secondary components are removed at about 20 milliTesla (mT) and a few specimens from each site were therefore subjected to AFD. Orthogonal vector plots (Zijderveld, 1967) of selected specimens are shown in Fig. 4. In Fig. 4D AFD behaviour of specimen NLP 3-1a shows a secondary magnetization was removed in a field of 5 mT. and the direction remains stable during further demagnetization. Figures 4A, B and C illustrate how thermal demagnetization is helpful in removing the secondary components at lower temperatures before the original magnetization, possibly acquired at the time of rock formation, is revealed at higher temperatures as a convergent component. Specimens from the NL also show stable directions at higher temperatures with thermal demagnetization and with AFD. One important observation to emerge is that the specimens from the same sample and samples of the same site, which show consistent preliminary directions, also show consistent and stable directions following both demagnetization tests. Although some specimens from other sites exhibit stable behaviour there is lack of coherence of their directions even following demagnetization tests.

The results of the demagnetization (both thermal and AFD) show that the ALR site from the KB has both stable normal and reversed directions. Sites NLP, NDC, RKT and BNG have stable, and predominantly normal, directions although some specimens in each site show deviations. In




Sampling details of Banganapalli Quartzite and Narji Limestone in Palnad Basin

| Site No. | Site | No. of Samples | No. of specimens |
|---|---|---|---|
| **Banganapalli Quartzite** | | | |
| 1. | Jaggayyapeta (JP) | 11 | 19 |
| 2. | Annavaram (ANV) | 2 | 4 |
| 3. | Hemadri (Vedadri - HMD) | 10 | 15 |
| 4. | Medlacheruvu (MCR&MLC) | 14 | 23 |
| 5. | Kandibanda (KNB) | 8 | 16 |
| 6. | Choutapalli (CTP) | 3 | 6 |
| 7. | Gurrambodu Tanda (GBD) | 5 | 10 |
| 8. | Rajavaram (RJV) | 3 | 5 |
| 9. | Nagarjunasagar (NSR) | 36 | 68 |
| 10. | Ethipotala (ETP) | 2 | 4 |
| 11. | Veerlapalem (VRP) | 3 | 3 |
| | **Total** | **97** | **173** |
| **Narji Limestone** | | | |
| 11. | Veerlapalem (VRP) | 11 | 27 |
| 12. | Damaracherla (DMC) | 4 | 9 |
| 13. | Tallaveerannagudem (TVG) | 4 | 7 |
| | **Total** | **19** | **43** |

**Fig.3.** Outcrops of Banganapalli Quartzite and Narji Limestone in Palnad basin. *Inset:* Outline map of Cuddapah basin showing the area sampled for palaeomagnetic study. Details of sampling are given in table as inset. Numbers in the Figure indicate site numbers given in table. F-F: Fault. (*after* Geological Survey of India, 1981).

the HMD site of the PB all specimens show a stable reversed direction. From the same basin, sites MCR and MLC and NSR show stable normal and reversed directions. Other sites from the PB show either normal or reversed directions in a few specimens at each site while the remaining specimens do not conform to any pattern. VRP and DMC sites of the NL have stable normal directions. All demagnetized directions from 173 specimens (54 samples) of BQ from both the KB and PB are tilt-corrected and the sample means are shown in Fig.5 along with the mean normal and reversed directions. The sample means from NL from 15 specimens (7 samples) are shown in Fig.6 along with the mean normal magnetization direction. When normal and reversed directions occur in the same stratigraphic layer such as in BQ (Fig. 5), the Characteristic Remanent Magnetization (ChRM) may be used for a Reversal test (Butler, 1992). This indicates that the ChRM directions are free of secondary components of magnetization. The sampling of BQ is restricted to the gritty zone and the brown coloured zone which are together restricted to a thickness ranging between 5 and 20m. Further, the reversal test is applied to the mean normal and reversed directions shown in Fig.5. The directions deviate by 7.8°. Piper and Crowley (1999) consider the angular difference of 7.5°, between normal and reversed directions in the Peel Sandstone of 410-400 Ma, is less than the critical angle of 12.1° and therefore yields a positive reversal test. Van der Voo (1990) and McFadden and McElhinny (1990) have given criteria for the reversal test, which have been applied to BQ and give no grounds for doubting that the two populations are antiparallel and imply that the magnetization of BQ is a ChRM acquired close to the time of sedimentation.

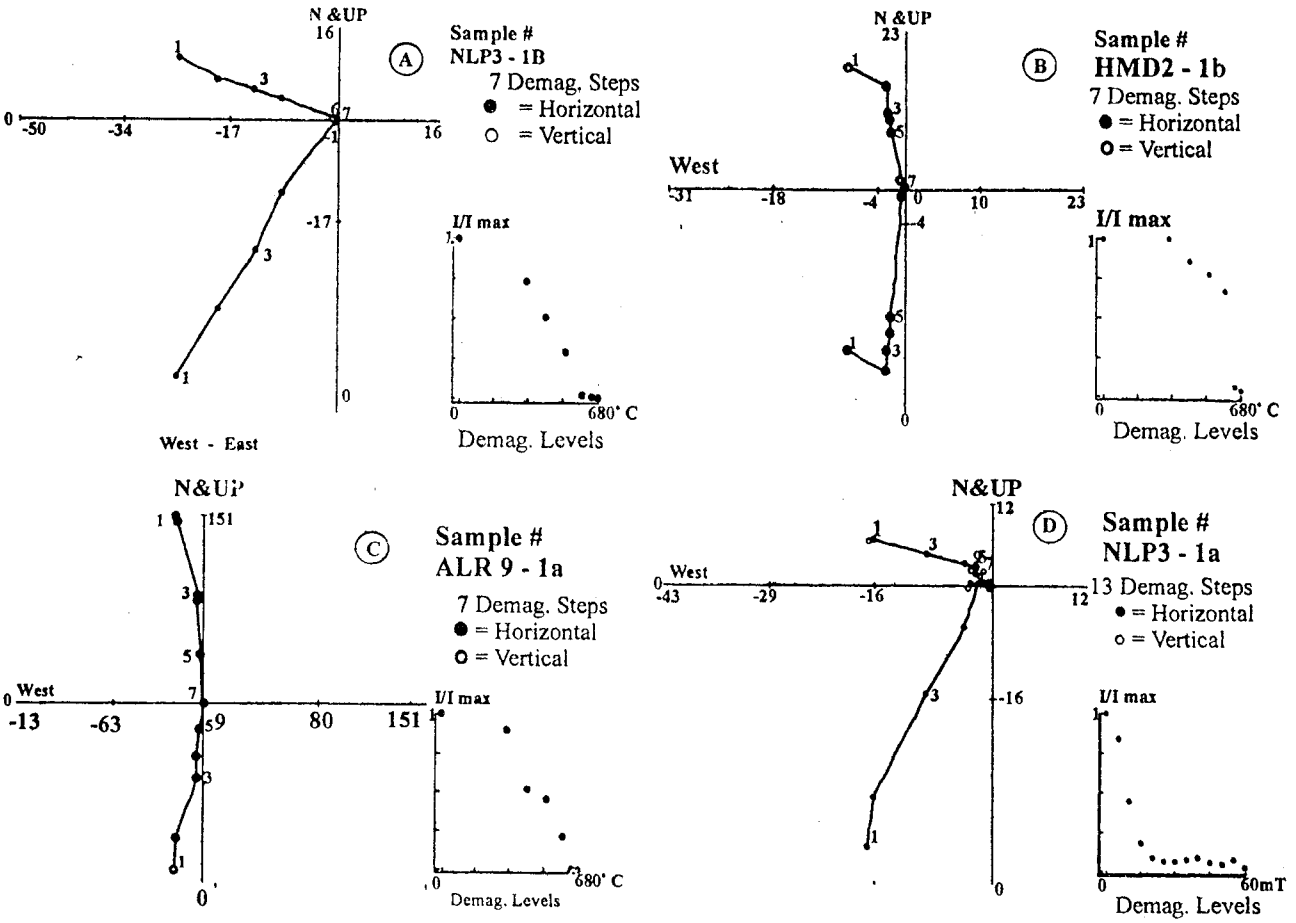



**Fig.4.** Orthogonal vector (Zijderveld) plots for representative Banganapalli Quartzite samples. Samples NLP3-1b (Fig.4A), HMD2-1b (Fig.4B), ALR9-1a (Fig. 4C) are thermally demagnetized and sample NLP3-1a (Fig.4D) is Af demagnetized. Small graphs at the right side of Figs. 4A, 4B and 4C shows fall of intensity with increasing temperature and in Fig.4D. with applied field.

Palaeomagnetic pole positions for BQ are calculated for both stable normal and reversed populations and also as a single population. The mean direction and pole position with other statistical parameters are given in Table 2 for BQ and NL. The mean directions are calculated from the sample means shown in Figs. 5 and 6.

## DISCUSSION

### Stratigraphic Correlation

In the geology of India the Kurnool Group of formations is considered to be stratigraphically equivalent to the Upper Vindhyan formations of central India. To correlate the palaeomagnetic data of the Kurnool Group with the Upper Vindhyan formations the available palaeomagnetic data and age information are now considered. The earliest attempt to determine a radiometric age from the Cuddapah basin was by Aswathanarayana (1962) who derived a model age for the Zangamrajupalle galena of 1400 Ma.

According to Aswathanarayana (1964) Cuddapah sedimentation began around 1080 Ma. However, based on Rb-Sr dating of sills near Pulivendla, yielding ages of 980 and 1359 Ma, Crawford and Compston (1973) consider that Cuddapah sedimentation began around 1700 Ma. According to Bhattacharji (1987) magmatism in the Cuddapah basin ranges from 1700-1000 Ma. Bhaskar Rao et al. (1995) determined the Rb-Sr age of a mafic sill near Pulivendla at 1800 Ma. For the Vempalle Limestone (Table 1), Zacharaiah et al. (1999) determined a Pb-Pb age of 1800 Ma. Recently Anand et al. (2003) gave Ar-Ar data of 1900 Ma for the sill intruding the Tadpatri Formation (Table 1). This means that the sedimentation in the Cuddapah basin may have begun earlier than about 2000 Ma. Based on the small digitate stromatolites, the Vempalle Formation is considered to belong to the upper part of the Palaeoproterozoic by Sharma and Shukla (1999). Therefore the Cuddapah sedimentation probably began earlier than 2000 Ma, which is well within the upper part of Palaeoproterozoic. This conclusion is

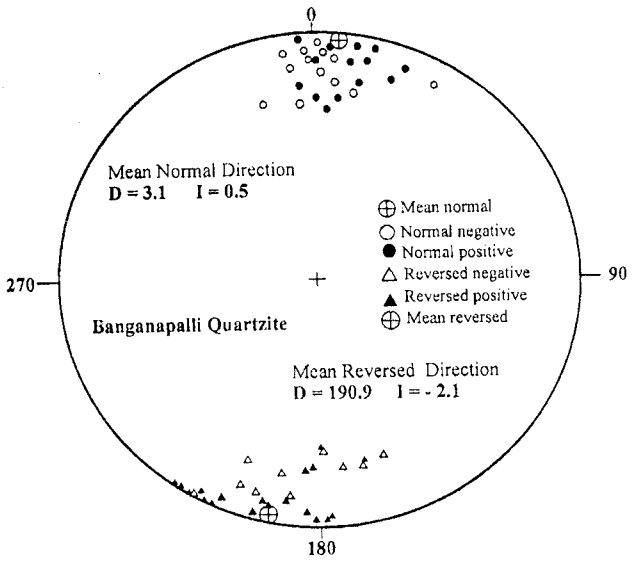



**Fig.5.** Equal area projection- lower hemisphere showing tilt-corrected thermal and Af demagnetization results along with mean normal and reversed directions of Banganapalli Quartzite. Points indicate sample means.

important for any correlation of palaeomagnetic directions from the formations of Cuddapah Supergroup and Kurnool Group with the Vindhyan Supergroup.

The Vindhyan Supergroup comprises the other Proterozoic sedimentary formations in India from which considerable palaeomagnetic data are available. Reviewing geochronological work on the Vindhyan Supergroup, Williams and Schmidt (1996) suggest that the base is at least 1200 Ma in age and could perhaps be as old as 1400 Ma.

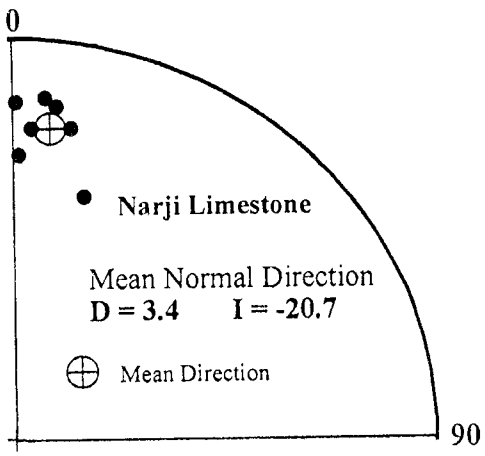



**Fig.6.** Equal area projection- lower hemisphere showing tilt-corrected thermal and Af demagnetization results of Narji Limestone along with mean direction. Points indicate sample means.

They also suggest that the Kaimur Group sedimentation may have begun about 1150 Ma and that the upper Rewa and Bhander Formations belong to the Neoproterozoic. Kumar et al. (2001), based on the Rb-Sr ages of the glauconites from a lower Semri Formation of the lower Vindhyan, have shown that Vindhyan sedimentation began about 1600 Ma ago. Similar U-Pb zircon ages of 1628 and 1631Ma have been reported from the lower Semri Formation by Rasmussen et al. (2002) and Ray et al. (2002) respectively. Sharma and Shukla (1999) find biostratigraphic marker fossils in the Owk Shale of the Kurnool Group of formations and in the Rewa and Bhander Groups of the Vindhyan Supergroup to be similar and consider that they belong to the Neoproterozoic. According to Ray et al. (2002) the strontium isotope ratio from the Bhander Limestone suggests a Neoproterozoic age. Simplified magnetostratigraphic columns for the Cuddapah Supergroup and the Kurnool Group, together with those of Vindhyan Supergroup and the Proterozoic of Siberia are shown in Fig.7. The stratigraphy shown in Fig.7 is not to scale, in all cases, and is only diagrammatic for the purposes of comparison.

Krishnan (1982) considered that a diamondiferous conglomerate separates upper Rewa Group of the Vindhyan Supergroup from the Lower Kaimur Group of formations. This has led to the classic correlation (Krishnan, 1982) of the Kurnool group of formations in the Cuddapah basin with the Rewa and Bhander groups of formations in the Vindhyan basin. It has been based on the presence of the diamond bearing Banganapalli Quartzite (conglomeratic zone) as a basal formation in the Kurnool Group. Chaudhuri et al. (1999), attempting a chronostratigraphic classification of Proterozoic formations of India, also correlated the Kurnool Group with the Rewa and Bhander Groups in the Vindhyan Supergroup based on the diamond bearing conglomerates at their base. However, Soni et al. (1987) in a review of the Vindhyan Supergroup consider that the diamondiferous conglomerate is not at the base of the Rewa Group but is an intraformational conglomerate at the top of the Lower Rewa Sandstone (or Itwa Sandstone), which is below the Jhiri Shale (Table 3). According to these authors there is also another diamondiferous conglomerate in the Vindhyan formations at the base of the Ganurgarh Shale of the Bhander Group. They also mention that there are two diamondiferous kimberlite diatrems emplaced in the Baghain Sandstone of the Kaimur Group of the Vindhyans. In view of the above discussion and by following the classical methods of geological correlation, if an attempt is made to correlate the Kurnool Group with the Rewa and Bhander Groups of the Vindhyan Supergroup based on the occurrence of diamondiferous conglomerates, the lower portion of the

**Table 2.** Mean directions, palaeomagnetic poles with statistical parameters and palaeolatitudes of Banganapalli Quartzite, Narji Limestone and Baghain Sandstone

| | S | N | D° | I° | $\lambda_s$ | $\phi_p$ | $\lambda_p$ | $\phi_p$ | $\alpha_{95}$ | k | Palaeo. Lat* |
|---|---|---|---|---|---|---|---|---|---|---|---|
| NL | 2 | 7 | 3.4 | -20.7 | 16.7 | 79.80 | 62.4 N | 252.6 E | 7.1 | 73.6 | 6.39 S |
| BQ Normal | 10 | 27 | 3.1 | 0.5 | 15.7 | 79.05 | 74.57 N | 247.33 E | 6.7 | 18.1 | 5.5 N |
| BQ Reversed | 7 | 27 | 190.9 | -2.1 | 15.7 | 79.05 | 72.04 S | 41.21 E | 8.0 | 13.1 | 4.2 N |
| BQ mean | 13 | 54 | 7.0 | 1.3 | 15.7 | 79.05 | 73.48 N | 233.63 E | 5.2 | 15.0 | 5.95 N |
| BS Gr. 'A' Normal | 45 | 18 | 5.0 | 39 | 24.7 | 80.18 | 84.85 N | 196.01 E | 10.5 | 53.9 | 18.3N |
| BS Gr. 'A' Reversed | 36 | 16 | 179.4 | -38 | 24.7 | 80.18 | 86.85 S | 90.33 E | 11.7 | 43.8 | 25.0 S |
| BS Gr. 'B' Normal | 16 | 7 | 347.0 | -40 | 24.7 | 80.18 | 41.15 N | 276.1 E | 31.9 | 63.4 | 26.46 S |
| BS Gr. 'B' Reversed | 22 | 17 | 179.0 | 43.5 | 24.7 | 80.18 | 40.16 S | 81.29 E | 17.3 | 29.1 | 21.68 N |
| Mean of Gr 'A' | 81 | 34 | 1.9 | 39.0 | 24.7 | 80.18 | 72.0 N | 340.2 E | 6.2 | 51.3 | 18.4 N |
| Mean of Gr 'B' | 38 | 24 | 354.9 | -43.0 | 24.7 | 80.18 | 40.1 N | 266.2 E | 9.5 | 36.6 | 28.4 S |

NL= Narji Limestone; BQ=Banganapalli Quartzite; BS= Baghain Sandstone; S=No. of sites; N= No. of samples; D= Mean declination; I= Mean inclination; $\lambda_s$ and $\phi_s$ = site latitude and longitude; $\lambda_p$ and $\phi_p$ are VGP latitude and longitude; $\alpha_{95}$ = 95% circle of confidence; k = precision parameter; * Palaeolatitude with reference to Nagpur

Banganapalli Quartzite can be correlated either with the upper portion of Lower Rewa Sandstone (Model 1) or the lower portion of the Ganurgarh Shale (Model 2) as shown in Table 3. There is no particular preference to either of these two models. However, it is possible that palaeomagnetic studies on the Banganapalli Quartzite could resolve the issue.

Poornachandra Rao and Bhalla (1996) reviewed the palaeomagnetic work on the Vindhyan Supergroup and produced a Geomagnetic Polarity Time Scale (GPTS) for the period of Vindhyan sedimentation. This was revised by Poornachandra Rao et al. (1997) by adding a reversed zone in the Kaimur Group representing the magnetization of the Baghain Sandstone, which actually contains both normal and reversed directions. The revised GPTS is called the Vindhyan Geomagnetic Polarity Time Scale (VGPTS). A GPTS from Siberian rocks (SGPTS) for the Riphean period has been given by Kiramov (1987). These GPTSs are shown in Fig.7 and arranged in the Figure in such a way that the base of VGPTS at 1600Ma matches the corresponding age on the SGPTS. The Cuddapah-Kurnool GPTS (CKGPTS), with a base at 2000 Ma (Anand et al. 2003) is also shown. There is a comparatively better age control of the VGPTS than the CKGPTS. Of course, the SGPTS has more reversals with better palaeomagnetic age control and is currently the best GPTS available from Proterozoic rocks anywhere in the world.

Reviewing the palaeomagnetic work on the Cuddapah basin, it is clear that the CKGPTS consists of only one R zone, which is recorded in the Srisailam Formation. All the formations belonging to the lower groups, namely the Papaghni, Chitravati and Nallamalai, are considered as N since only N results have been obtained from the Gulcheru, Tadpatri and Bairenkonda Formations (Prasad et al. 1987). There are many formations within the different groups of the Cuddapah Supergroup (Table 1). Out of them only the Gulcheru, Tadpatri and Bairenkonda Formations have so far been studied. Based on this work it is considered that the Cuddapah Supergroup is generally of normal magnetization except for the uppermost Srisailam Formation, which is of reversed magnetization.

The Banganapalli Quartzite (BQ) has given both R and N directions while the Srisailam Quartzite below is reversely magnetized with D/I=149°/0°, and the Narji Limestone immediately above it is normally magnetized with D/I = 3.4°/-20.7°. This sequence with a formation showing both R and N occurring between those showing only R and N directions possibly suggests that the Banganapalli Quartzite was being deposited when the Earth's magnetic field was undergoing more rapid reversal. The Baghain Sandstone (BS) in the VGPTS also shows both R and N polarities. Hence two similar formations separated by a large distance of about 1000 km (Fig.1, inset map) could record the same changes of geomagnetic field polarity. Such polarity transitions take place within short geological time of about less than 10,000 years (McElhinny and McFadden, 2000). Therefore the deposition of BQ and BS may have been very rapid. This suggests the possibility of stratigraphically correlating two formations one (BQ) at the base of the Kurnool Group and the other (BS) from the Upper Kaimurs of the Vindhyan Basin (Fig.7) and implies that they are nearly coeval.

The correlation of CKGPTS and VGPTS is fraught with many difficulties. Firstly, there is a difference of 400 Ma between the beginning of the Cuddapah and Kurnool formations and the Vindhyan formations with the former

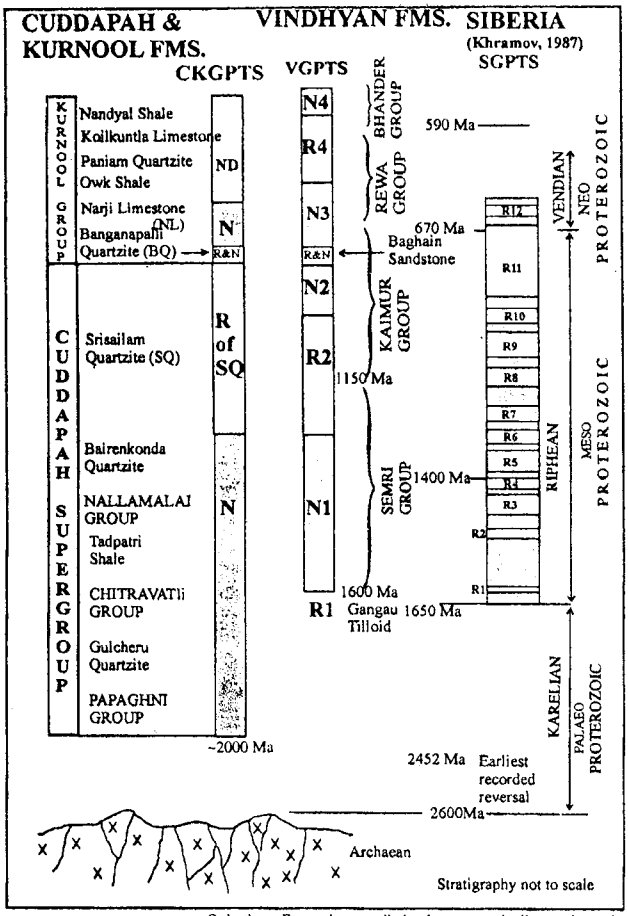



**Fig.7.** Possible Stratigraphic correlations based on palaeomagnetism. CKGPTS (Cuddapah-Kurnool Geomagnetic Polarity Time Scale); VGPTS (Vindhyan Geomagnetic Polarity Time Scale); SGPTS (Siberian Geomagnetic Polarity Time Scale-Khramov, 1987).

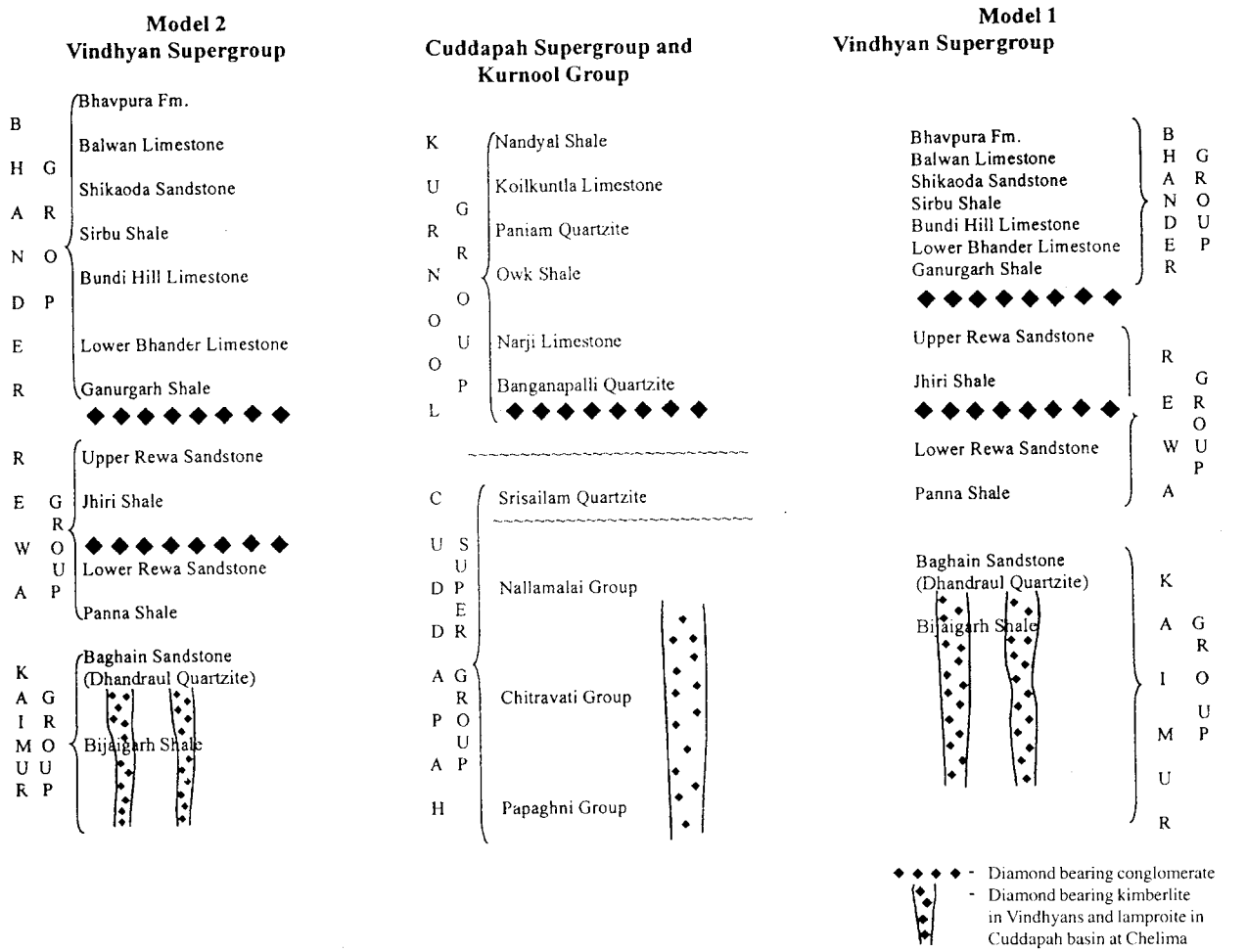
being older. Secondly, there are only two reversals so far recorded in the Cuddapahs and Kurnools while there are five reversals found in the Vindhyans. Certainly the record of reversals in these Proterozoic formations is very few compared to those in the SGPTS (Fig.7). This could mean either the sedimentation is very fast in the two formations (faster in Cuddapah and Kurnools than in the Vindhyans) or more reversals may be discovered in them in future. If rapid deposition took place the available radiometric ages cannot be reconciled and the correlations using biostratigraphic marker horizons cannot be disputed. However, an attempt is made to correlate the two formations (Cuddapah and Kurnools and Vindhyans) based on the available palaeomagnetic data. The acronyms CKGPTS and VGPTS are used only to identify one from the other in Fig.7 and it does not mean that the reversal record in each of the stratigraphic sequences is established and that the polarity time scales are given.

As shown in Fig. 7, the Cuddapah Supergroup N could be correlated with the N1 of the Semri Group, and the R zone of the Srisailam Quartzite (SQ) could possibly correlate with R2 of the VGPTS. This R zone in SQ is shown continuing to the top of N2 in the VGPTS. However, it could be shorter because of a hiatus between the Cuddapah Supergroup and the Kurnool Group (Table 1). Since both N and R directions are encountered in the Banganapalli Quartzite (BQ) it may be equated with a similar formation, the Baghain Sandstone, from which both R and N are reported (Poornachandra Rao et al.1997). The palaeomagnetic records in BQ and the Baghain Sandstone (BS) are very similar with both positive and negative inclinations of R and N directions. The similarity ends there. Poornachandra Rao et al. (1997) reported two reversals in A and B groups of directions of BS. The 'A' group has normal and reversed magnetizations with positive and negative inclinations respectively while group 'B' show vice versa. The N2 of VGPTS (Fig.7) has a direction around D/I= 360°/32° (Poornachandra Rao and Bhalla, 1996). If a reversed direction of this N2 is recorded in the BS within the same Kaimur Group of formations, it should show D around 180° and I should be negative whatever its value. This condition is satisfied by the Group 'A' directions in the Baghain Sandstone. There is a normal positive direction within the same group which shows that the Baghain Sandstone has recorded a magnetization when the Earth's magnetic field was undergoing a polarity change as in the case of BQ. But the record of "B' group of directions (R +ve and N –ve) within the same Baghain Sandstone may mean that India was simultaneously in the northern hemisphere ('A' group) as well as in southern hemisphere ('B' group) during the deposition of the sandstone, which is an improbable proposition (see palaeolatitudes in Table 2). Therefore the other alternative may be that the 'B' group of directions with N –ve and R +ve are likely to be an overprint because of their similarity to Deccan Trap directions in general (Sreenivasa Rao et al. 1985). Therefore considering only the group 'A' directions of the Baghain Sandstone (although the inclinations are high compared to BQ) it is considered that the two formations BQ and BS are coeval. In which case the N zone of the Narji Limestone (NL) could be coeval with N3 of the VGPTS. The revised VGPTS of the Vindhyans of Poornachandra Rao et al. (1997) is further slightly revised to show the 'A' group's R and N character of the Baghain Sandstone instead of the single R shown by the authors. This correlation has two problems. Firstly, the R1 zone of the VGPTS is not apparent in the Cuddapah

Supergroup (CSG) possibly because R1 was not recorded in the CSG due to lack of sedimentation as suggested by a number of unconformities in the CSG. Alternatively R1 may possibly show up in the Chitravati Group with further study. The other problem is with the N2 zone in the VGPTS, which is not recorded in the CSG. The explanation is simpler here because the contact between the BQ and the Srisailam Quartzite is an unconformity between two Groups of formations, which is possibly of longer duration and is marked by a conglomerate bed at the base of the BQ. The N2 zone of the VGPTS was possibly missed in the CSG during this hiatus. This correlation can be made because it is considered that the cratonisation of India to form a single tectonic block occurred before intrusion of the dyke swarms into the basement of the Cuddapah and Vindhyan basins (Damodara Reddy et al. 1995).

Palaeomagnetic correlations discussed above indicate that the Banganapalli Quartzite and Narji Limestone correlate approximately with the upper part of the Kaimur Group. This correlation as shown in Fig.7 makes the Kurnool Group slightly older than formerly believed with its base possibly equivalent to the upper part of Kaimur Group of the Vindhyan Supergroup. This correlation is different from both models based on the geological correlation. This correlation of diamondiferous conglomerates (Table 3) may be inappropriate because there are diamond-bearing kimberlites intruding up to the level of the Baghain Sandstone (Soni et al.1987) whilst the diamondiferous conglomerate are found at a slightly higher stratigraphic level in Vindhyans. On the other hand no kimberlite or lamproite occurrences are known in the Kurnool formations. The source for diamonds in the

**Table 3.** Possible geological correlations of Kurnool Group (Cuddapah basin) with Rewa and Bhander Groups (Vindhyan basin) based on diamondiferous horizon. Stratigraphically of Cuddpah Supergroup *after* Nagaraja Rao et al. 1987 and stratigraphy of Vindhyan Supergroup *after* Soni et al. 1987



Banganapalli Quartzite of Kurnool basin (BQ of the Palnad basin is not diamondiferous) is considered to be the Wajrakarur kimberlites (Vaidyanadhan, 1965; Chetty and Satyanarayana Rao, 1986) and/or Chelima lamproites (Babu, 1998), which are known to be diamondiferous. Wajrakarur kimberlites intrude into the granites on the western side of the Cuddapah basin (Fig. 1). The diamondiferous lamproites at Chelima (diamondiferous nature of Zangamrajupalle lamproite is not known) intrude into the Cumbum sediments (Tables 1 and 3 and Fig. 1) and occur outside the Kurnool basin on its eastern margin (Appavadhanulu, 1966; Bhaskara Rao, 1976; Bergman and Baker, 1984; Babu, 1998), while the diamondiferous BQ is on the western margin of the Kurnool basin. Old workings for diamonds are found at Chelima but the lamproite at Zangamrajupalle is found in drill holes (Babu, 1998).

The Krishna lamproite field (KLF) in the Palnad basin (Fig.1) consists of different kinds of lamproites intruding into the granite-gneiss rocks close to the northeastern horn shaped peripheral portion of the Cuddapah basin (Reddy et al. 2003), but the BQ of the Palnad basin is not diamondiferous. However, several large diamonds to smaller ones are being recovered from a belt of gravels and alluvium distributed over a zone of width of about 20 to 30 km from the eastern margin of the northern tip of the Cuddapah basin. Therefore the correlation of Cuddapahs and Kurnools to the Vindhyans based on the diamondiferous zones may not be appropriate since the nature of the occurrence in the two basins is not similar.

The correlation of the Banganapalli Quartzite of the Kurnool Group (in the Cuddapah basin) with the Baghain Sandstone in the Kaimur Group (Vindhyan basin) is based on magnetic polarities involves correlation of two geologically similar formations of comparable age separated by about 1000 kilometers. Correlation of the VGPTS or CKGPTS with the SGPTS is not possible because of many identified reversals in SGPTS. The palaeomagnetic poles of BQ (for N, R and mean of N and R), NL and BS (for N and R of group 'A', N and R of group 'B', mean 'A' and mean 'B') are shown in Table 2 and Fig. 8. The closeness of BQ poles and their mean pole to group 'A' poles and the mean pole of BS strengthens the correlation of BQ with BS. As suggested earlier the group 'B' poles of BS are far away from both BQ poles and group 'A' poles of BS and may indicate a remagnetization at some later time.

**Formation of Gondwana**

The palaeolatitude of India, with reference to Nagpur deduced from the Banganapalli Quartzite is calculated to be 6.2° N and for Narji Limestone it is 5.1° S (Table 2). Palaeolatitudes from all Proterozoic formations of India studied palaeomagnetically range between 28.4° S to 31.0° N (Poornachandra Rao et al. 1997; Prasad et al. 1987) and suggest that India was situated in equatorial to tropical latitudes during these times. Red beds consisting of conglomerates and arkoses (such as those of the Banganapalli Quartzite) and sandstones and shales containing hematite as a visible pigmenting material, are prominent clastic formations of the Proterozoic Eon (Windley, 1995). In India the Cuddapah, Kurnool (Subba Reddy and Prasad, 1999) and Vindhyan (Soni et al. 1987) groups of formations contain red beds. These rocks appear to have formed under warm moist (tropical) as well as hot dry (desert) conditions. Evidence for similar climatic conditions is available from palaeomagnetism (low palaeolatitudes) of the formations which are regarded as diagnostic of warm conditions from the Mesoproterozoic McArthur basin in Australia (Idnurm and Giddings, 1988; Idnurm et al 1995). A low latitudinal position for Australia in Neoproterozoic (about 650 – 590 Ma) also comes from red beds of the Adelaide fold belt (Schmidt and Williams, 1995, 1996). According to Salop (1983) the Bangemall

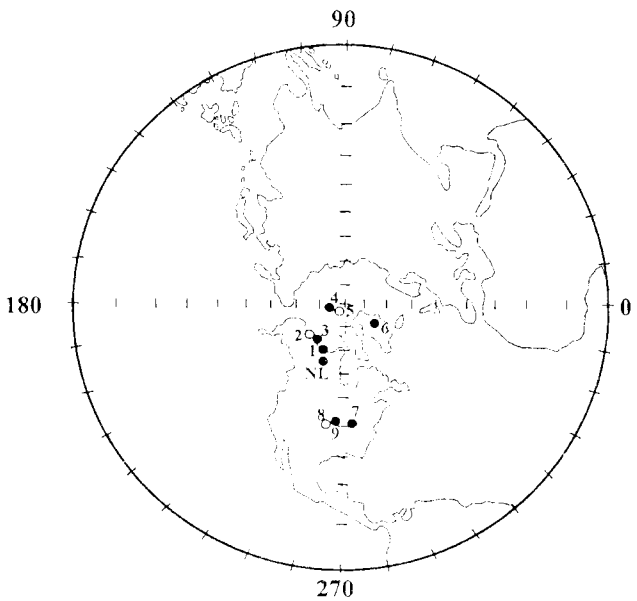



**Fig. 8.** Palaeomagnetic poles of Banganapalli Quartzite (BQ), Narji Limestone (NL) (Kurnool Group) and Baghain Sandstone (BS- Kaimur Group). Equal area North Polar Projection. NL= Narji Limestone; (1) BQ Normal magnetization, (2) BQ Reversed magnetization, (3) BQ mean direction, (4) BS group 'A' normal magnetization, (5) BS group 'A' reversed magnetization, (6) Mean magnetic direction of BS group 'A', (7) BS group 'B' normal magnetization, (8) BS group 'B' reversed magnetization, (9) Mean magnetic direction of BS group 'B'.

basin in western Australia is the age equivalent of the Cuddapah and Kurnool formations containing similar rock formations, giving further evidence for the formation of red beds both in India and Australia contemporaneously at near equatorial latitudes. Thus the Proterozoic palaeomagnetism and palaeoclimatic information from India and Australia seem to agree well.

According to Karlstrom et al. (2001) the concept of a super continent cycle remains central for understanding the long-term evolution of the tectonism on the Earth and that the super continent Rodinia was assembled by continental collisions at about 1.0 Ga. Idnurm and Radhakrishna (1999) suggest that the APWP for India and Australia are separate during the Palaeoproterozoic but appear to merge towards the end of the Proterozoic and indicate that the two continents were amalgamated in their Gondwana relative position either by about 1200 Ma or between 1000 and 750 Ma. Harlan et al. (2003) and Karlstrom et al. (2001) suggested that about 755 Ma Laurentia and western Australia were rifting apart. If Australia was rifting from Laurentia, it is possible that it was moving towards India to give rise to Gondwana. Ahmed (*personal communication*, 1989) is also of the same opinion based on the palaeontological data. Radhakrishna (1999 a, b) discussed the amalgamation of UR (India-Sri Lanka- Western Australia) and NENA (central Australia-Antarctica–Laurentia) to form Rodinia as a precursor to Gondwana based on the excellent agreement of poles from Australia and India at 1080 Ma and a cluster of poles in a coherent narrow band from India, Australia, Antarctica and Africa (cratonic segments of Gondwana) at 550 Ma. Powell and Pisarevsky (2002) suggest that some continental block collided with western margin of Australia at about 1080 Ma. They also indicate that India could have been in contact with the western margin of Australia at 610 Ma and that the Darling mobile belt being a sinistral shear at that time, the northwest principal compression is consistent with an oblique collision of India with the margin of western Australia. Suryanarayana and Anjanappa (1975) made extensive joint pattern studies through contour diagrams and quartz fabric analysis of quartz grains in granite at the southern margin of the Cuddapah basin and suggested that the dyke swarms were intruded into a fracture system developed as a result of E-W compression. The western thrusting present on the eastern margin of the Cuddapah basin (Nagaraja Rao et al. 1987) giving it a crescent shape and western concave bending of the Eastern Ghats suggest that the thrusting of India came from the east. The trends and directions of large scale thrusting and flexuring across Pilbara, Hamesley and Bangemall basin of western Australia (Geological Society of Australia, 1971) indicate the possible collision with India along its eastern margin (Prasad et al. 1987).

Therefore palaeomagnetic data (palaeolatitudes and pole positions), palaeoclimates (based on the presence of red beds) palaeontological information and the presence of suggestive structures on both the continents indicate a collision of India and Australia in the formation of Gondwana possibly in late Neoproterozoic or early Palaeozoic.

## CONCLUSIONS

The main inference of this work is that the correlation between two distant groups (Kurnool and upper Vindhyan) based on the occurrence of diamondiferous formations may not be entirely correct since the sources of diamonds are dissimilar. On the other hand, palaeomagnetism, which is a global phenomenon, provides an alternative control over stratigraphic correlations. This is suggested by the presence of a reversal record in a narrow zone in both the Upper Vindhyan and Kurnool Groups. This also indicates that the possibly coeval narrow zones in the Kurnool (Banganapalli Quartzite) and in the upper Vindhyan (Baghain Sandstone) were formed at the time when the Earth's magnetic field was undergoing a polarity transition. The palaeomagnetic data together with palaeoclimatic and palaeontological information point to the occurrence of India and Australia in low latitudes during the Proterozoic and suggest a possible collision later to form the larger Gondwanaland.

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